SUPPORTING ADULT GUIDANCE

Finding an angle

Introduction to Trigonometry. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner finds an **angle** from two known sides using the **inverse ratios** $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$. The method mirrors finding a side: write the ratio the two sides make, substitute to get a value, then **apply the inverse** to get the angle — θ = ratio^{-1}(value). The emphasis is on **choosing the ratio from the two sides you have**, understanding the inverse as the operation that **undoes** sin/cos/tan, and using the calculator's inverse keys in **degrees**. The recurring error is giving the ratio value as if it were the angle (forgetting the inverse).

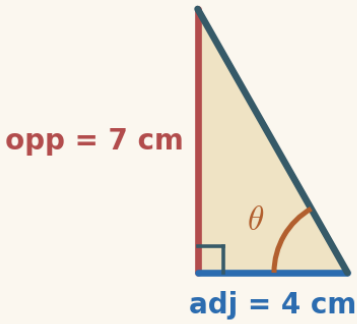
Driving Question: *How can two sides reveal the angle? (the inverse ratios)*

The mathematics, in plain terms

This lesson turns trigonometry around and, in doing so, teaches one of the most transferable ideas in mathematics: the **inverse**. Every operation that does something has a partner that undoes it — add/subtract, multiply/divide, square/root — and now sin and $\sin^{-1}$, cos and $\cos^{-1}$, tan and $\tan^{-1}$. Given two sides, the method is the familiar four lines with the inverse in place of the rearrangement. Take a triangle with opposite 3 and adjacent 4: **(1)** those two sides make the tangent, $\tan\theta$ = opposite/adjacent; **(2)** substitute, $\tan\theta = 3/4 = 0.75$; **(3)** apply the inverse to undo tan, $\theta = \tan^{-1}(0.75)$; **(4)** evaluate on the calculator in degrees, $\theta = 36.9°$. Two teaching points matter most. First, **choosing the ratio** now depends on *which two sides you have* (opposite & adjacent → tangent; opposite & hypotenuse → sine; adjacent & hypotenuse → cosine), not on a given angle. Second, the **inverse is essential**: the number 0.75 is the ratio, not the angle; the angle is $\tan^{-1}(0.75)$. The single commonest error is stopping at the ratio value, so it is worth naming out loud that the inverse is the step that produces an angle. The calculator must be in **degrees**, and learners need to find the $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$ keys (often a 'shift' or '2nd' of the ordinary sin/cos/tan). Framing the whole lesson as 'finding a side, run backwards' keeps it connected to what the learner already understands, rather than presenting a new method to memorise.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.



Example 1 — finding an angle with tangent

You know two sides — the opposite (7 cm) and the adjacent (4 cm) — and want the angle. Opposite and adjacent → tangent.

1. write the ratio	$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$
2. put in the numbers	$\tan\theta = \frac{7}{4} = 1.75$
3. apply the inverse (undo tan)	$\theta = \tan^{-1}(1.75)$
4. work it out on a calculator (degrees)	$\theta = 60.3°$

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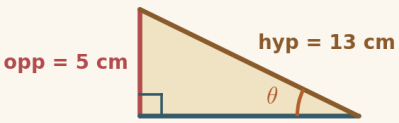

Example 2 — finding an angle with sine

This time you know the opposite (5 cm) and the hypotenuse (13 cm). Opposite and hypotenuse → sine.

1. write the ratio	$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$
2. put in the numbers	$\sin\theta = \frac{5}{13} = 0.385$
3. apply the inverse (undo sin)	$\theta = \sin^{-1}(0.385)$
4. work it out on a calculator (degrees)	$\theta = 22.6°$

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Every operation has an inverse

The heart of this lesson is the idea of an **inverse** — and it is worth making that idea explicit, because it recurs throughout mathematics. Adding is undone by subtracting; sin is undone by $\sin^{-1}$. Once a learner sees finding an angle as 'the reverse of finding a side', there is no new method to learn, only a new pair to recognise. Two supports matter. First, **choosing the ratio from the two sides**: name the two sides you have and pick the ratio that uses exactly those two — the interactives highlight them to make the choice visible. Second, the **inverse step**: insist that line 3 shows $\theta = \text{ratio}^{-1}(\text{value})$, and that line 4 is the calculator evaluation; separating them keeps the reasoning visible and prevents the 'ratio

value as angle' error. Mixing sine, cosine and tangent problems forces the learner to choose the ratio each time rather than pattern-matching. Finishing with real 'reveal the angle' problems (ramps, roofs, elevations) shows that two lengths you can measure hand you an angle you could not measure on the spot.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	From ratio back to angle: choose the ratio the two sides make, then reveal the inverse turning the ratio into the angle.	~10 min
Movement	Angle Finder: choose the ratio, type the inverse line $\theta = \text{ratio}^{-1}(\text{value})$, then evaluate on the calculator.	~14 min
Reflection	Spot the slip: catch the wrong ratio, and forgetting the inverse (giving the ratio value as the angle).	~10 min
Creativity	Reveal the angle: find a real angle (ramp, roof, elevation) from two measured lengths.	~6 min
Rest	A pause after seeing that the inverse simply runs the machine backwards.	~3 min
Nutrition	Mode B (intellectual): what the learning feeds — the idea of an inverse, and reusing a method in reverse.	~2 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Giving the ratio value as the angle (forgetting the inverse)

Repair: “0.75 is the ratio, not the angle. Apply the inverse: $\theta = \tan^{-1}(0.75) = 36.9°$.”

Choosing the wrong ratio for the two sides

Repair: “Which two sides do you have? Opposite & adjacent → tangent; opposite & hypotenuse → sine; adjacent & hypotenuse → cosine.”

Not finding the inverse keys on the calculator

Repair: “$\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$ are often the 'shift' or '2nd' of sin, cos, tan. Check $\sin^{-1}(0.5)$ gives 30°.”

Calculator in radians, not degrees

Repair: “Check the mode says DEG — $\sin^{-1}(0.5)$ should give 30, not 0.52.”

Rounding too early or forgetting the degree symbol

Repair: “Keep the full value until the end, then round as asked and write the angle in degrees.”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: SOH · CAH · TOA and choosing a ratio from two sides (Lesson 4), finding a side by writing and solving the ratio equation (Lessons 5–6), and using a calculator's inverse trig keys in degrees. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the learner choosing the ratio and applying the inverse, in that order:

- “Which two sides do you have? So which ratio do they make?”
- “Put in the numbers — what value does the ratio come to?”
- “That's the ratio, not the angle. What undoes tan? Apply $\tan^{-1}$.”
- “A ramp's rise and run — find the angle it makes with the ground.”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“What are $\sin^{-1}$, $\cos^{-1}$ and $\tan^{-1}$?”

They are the inverse ratios — the operations that undo sin, cos and tan. Where sin turns an angle into a ratio, $\sin^{-1}$ turns a ratio back into its angle. On the calculator they are usually the 'shift' or '2nd' of sin, cos, tan.

“How do I choose which ratio to use?”

Look at the two sides you have. Opposite and adjacent → tangent; opposite and hypotenuse → sine; adjacent and hypotenuse → cosine. Then the angle is that ratio's inverse of the value.

“Why isn't the answer just the ratio value?”

Because the ratio value (like 0.75) is a ratio of lengths, not an angle. You have to apply the inverse to convert it into an angle in degrees: $\theta = \tan^{-1}(0.75) = 36.9°$.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

NEO by Nudge Education · nudgeeducation.online · Curriculum by Gerry Docherty. Supporting Adult Guidance v0.1.